# Run report: Joux–Vitse $n = 5$ ladder and complete DLPs, 2026-10-15T05:18Z

Descriptive record of the runs (observations only; interpretation is in `../../README.md` and `../../REPORT.pdf`).

**Spec.** `PROTOCOL.md` (frozen, commit `ffccaf115`), orchestration `run.sh {ladder,dlp11,dlp31}`, binary `examples/joux_vitse_ladder.rs` at commit `1e8592da0` (rustfmt only after the freeze). Host: Claude Code cloud container, Intel Xeon @ 2.10 GHz, 4 vCPU, 15 GB, rustc 1.97.0; isolation none; the three orchestrations ran concurrently on the 4 vCPUs.

**Timeline.** First launch 04:33Z, killed by a container restart before 05:18Z (partial records in `data_killed_20261015T0518Z/`, not used). Relaunched 05:18Z. DLP `p = 11` seeds 1 and 2 completed 07:33Z and 11:02Z; seed 3 stopped by hand at 12:14Z with 0 relations (its curve has a one-point factor base; `data/dlp/p11_s3.STOPPED.md`). DLP `p = 31` completed 12:09Z (inside the 10-hour limit). Ladder completed 15:07Z.

**Requested and completed.** Ladder: 12 primes × 2 seeds × (32 F4 trials + `500` `p` meet-in-the-middle trials), all 24 files complete. DLP: 4 arms requested, 3 completed and verified, 1 stopped. Added after the freeze (addendum): one re-run of the committed `jv_quintic` plain test at `p = 271`, completed.

**Measured (unit: `F_p` multiplications; counts, not wall time).**

| quantity | value | n | spread | record |
|---|---|---|---|---|
| trial cost, all rungs | $5.9$–$9.2 \cdot 10^9$ (rung means) | 768 trials | min $4.6 \cdot 10^9$, max $3.3 \cdot 10^{10}$ | `data/ladder/*.jsonl` |
| trials with a decomposition | 491 of 1,422,768 | — | per-rung Wilson 95 % in `summary.json` | `data/ladder/` |
| F4 vs oracle disagreements | 0 | 768 | — | `f4_trial` rows, `agree` |
| relations failing to re-add | 0 | all | — | `relations_verified` |
| DLP `p = 11` s1: IC online / rho | $1.74 \cdot 10^{13}$ / $3.84 \cdot 10^5$ | 1 target | — | `data/dlp/p11_s1.jsonl` |
| DLP `p = 11` s2: IC online / rho | $2.67 \cdot 10^{13}$ / $1.52 \cdot 10^5$ | 1 target | — | `data/dlp/p11_s2.jsonl` |
| DLP `p = 31` s1: IC online / rho | $5.31 \cdot 10^{13}$ / $7.78 \cdot 10^5$ | 1 target | — | `data/dlp/p31_s1.jsonl` |
| `jv_quintic` plain test, today's engine | $1.104 \cdot 10^{10}$ | 3 random + 2 planted | — | `data/jv_quintic_rerun/` |

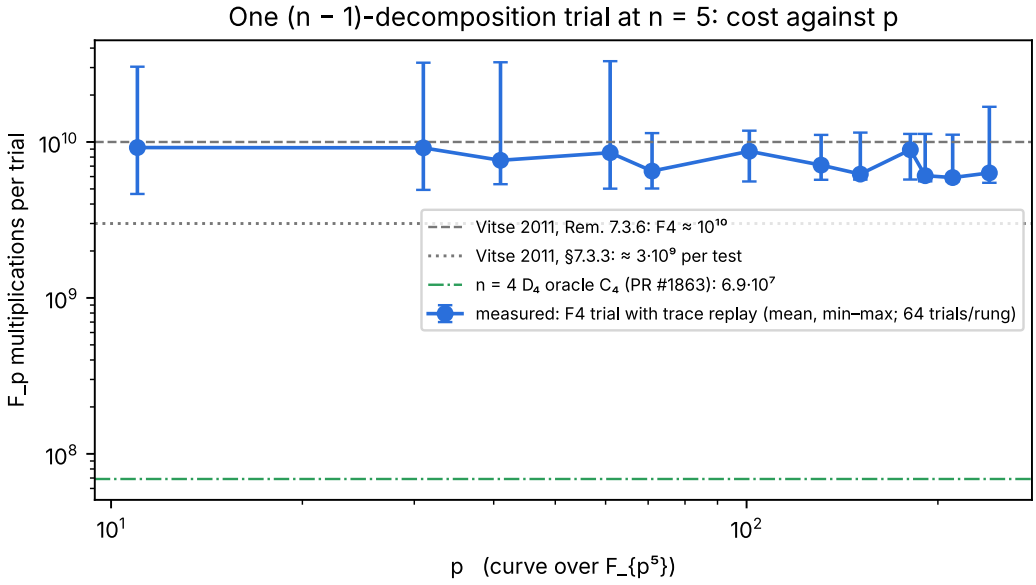



Figure 1: Cost of one trial against $p$ (mean, min–max; 64 trials per rung). Plotted from `summary.json` by `../../figures/plot.py`.

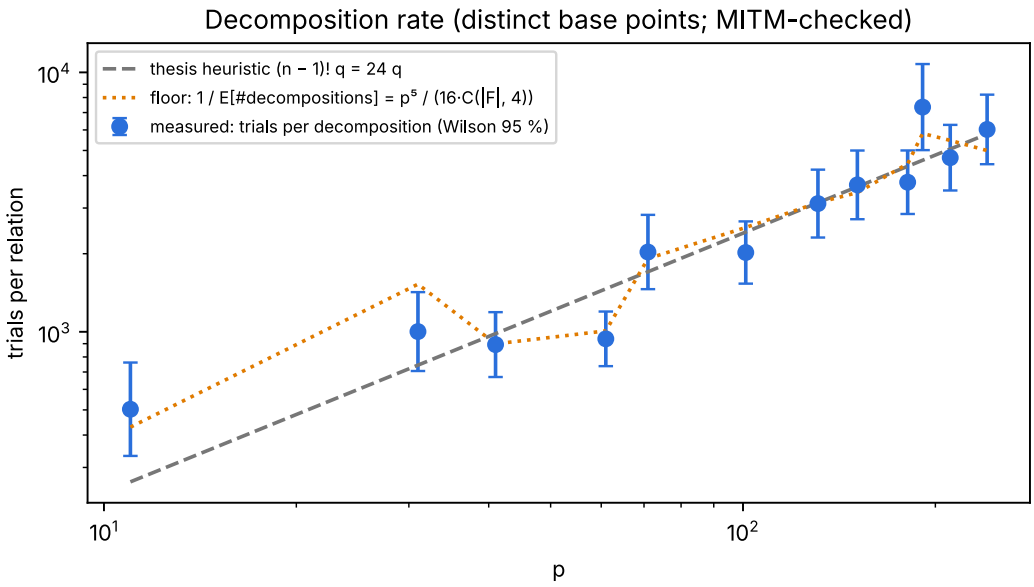



Figure 2: Trials per decomposition against $p$ (Wilson 95 %), from the same records.